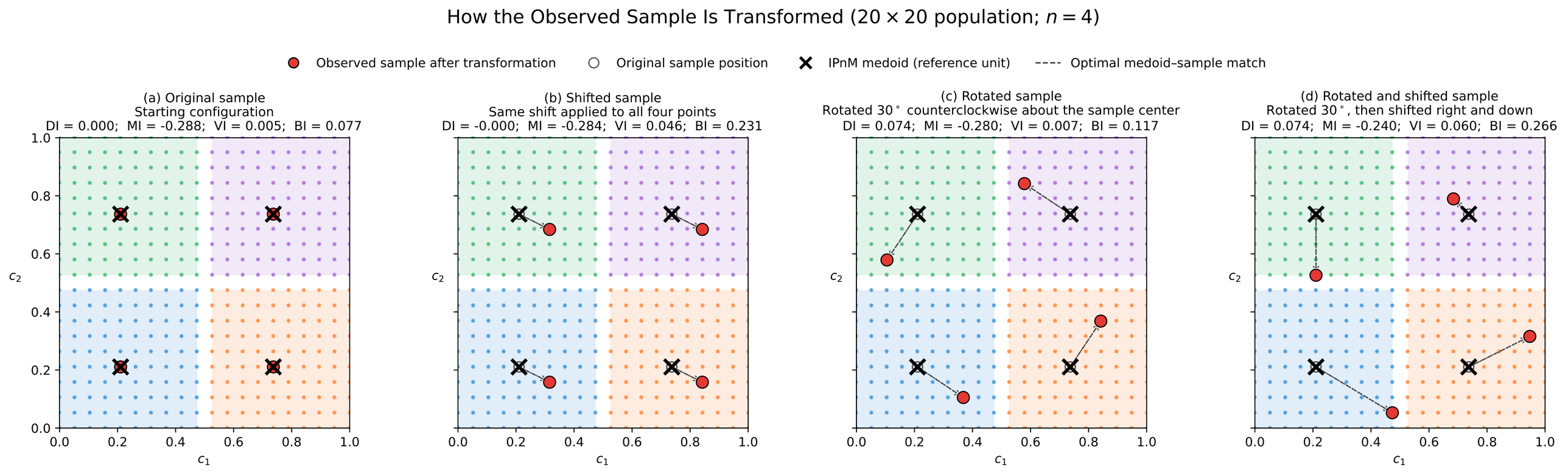

How the Observed Sample Is Transformed (20 × 20 population; $n = 4$)
Observed sample after transformation
Original sample position
IPnM medoid (reference unit)
Optimal medoid–sample match
(a) Original sample
Starting configuration
DI = 0.000; MI = -0.288; VI = 0.005; BI = 0.077
(b) Shifted sample
Same shift applied to all four points
DI = -0.000; MI = -0.284; VI = 0.046; BI = 0.231
(c) Rotated sample
Rotated 30° counterclockwise about the sample center
DI = 0.074; MI = -0.280; VI = 0.007; BI = 0.117
(d) Rotated and shifted sample
Rotated 30°, then shifted right and down
DI = 0.074; MI = -0.240; VI = 0.060; BI = 0.266
c_1
c_2
0.0
0.2
0.4
0.6
0.8
1.0